Risk Stratification of Pulmonary Embolism through Pulmonary Vascular Modeling

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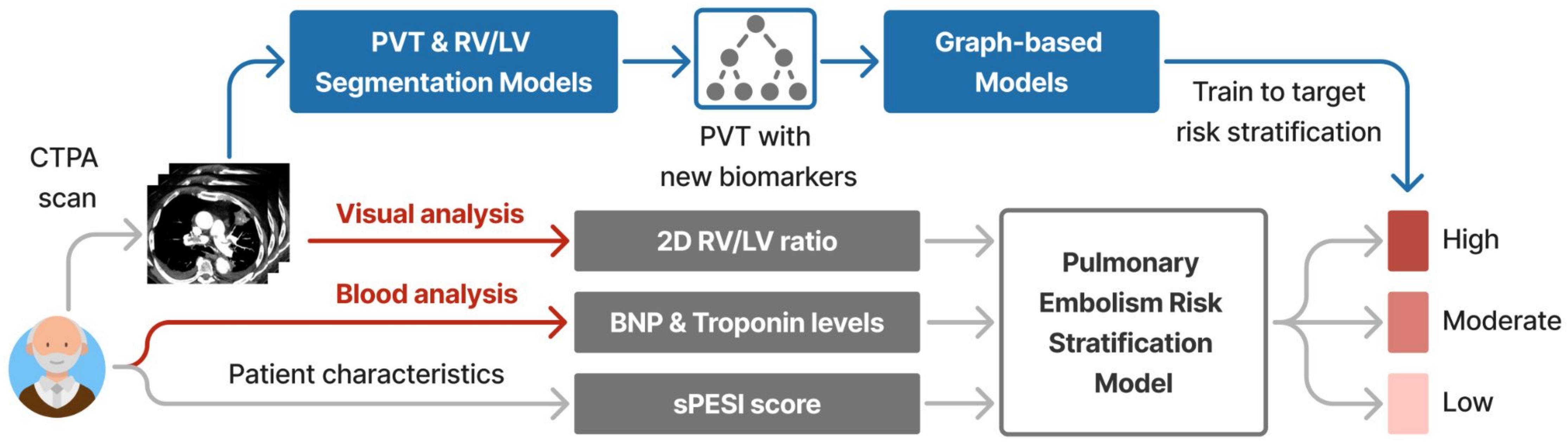
Goals

Enhance pulmonary embolism risk stratification accuracy and accelerate patient management.

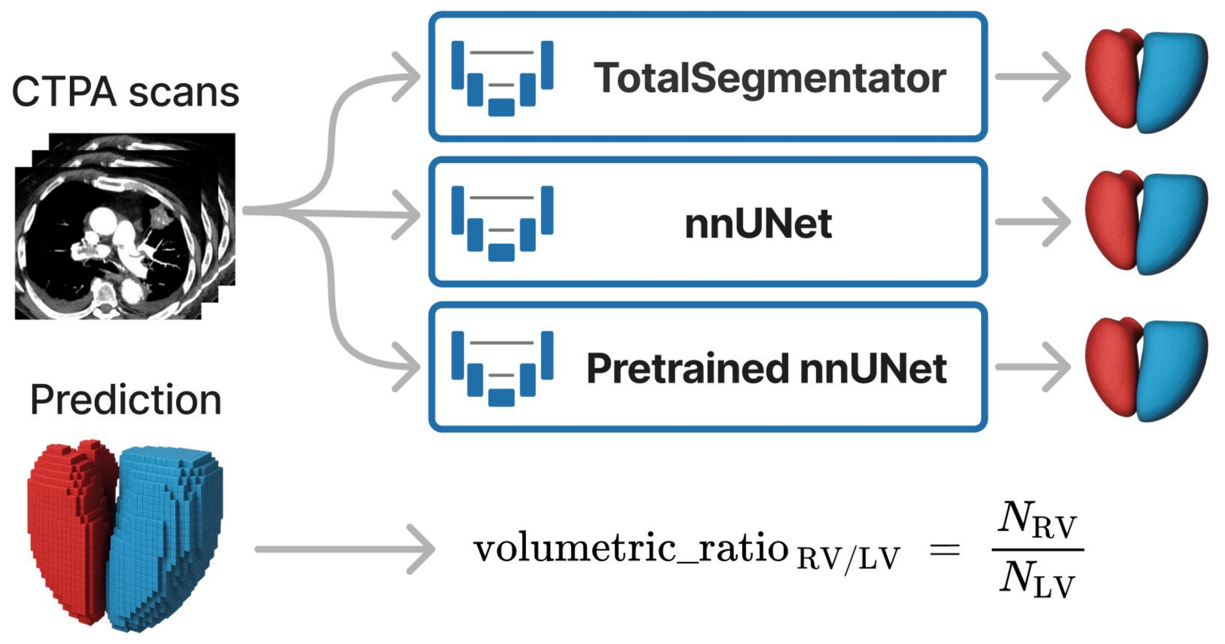
- **Automate segmentation of right and left ventricles** (RV, LV) to measure precise 3D RV/LV ratio
- **Pulmonary vascular tree (PVT)** saved as graph by segmenting arteries, embolies, and lung anatomy
- **Extract new local and global morphological biomarkers** with good predictive value from the PVT
- **Develop new graph-based deep learning models** for risk stratification using the PVT graphs and biomarkers



1. Visual analysis is performed by clinicians on a specific 2D slice, which may not accurately reflect the 3D RV/LV ratio, leading to potential risk stratification errors.

2. Blood analysis of BNP and Troponin, biomarkers of cardiac stress, is not systematically performed in clinical practice due to time and resources constraints.

RV/LV Ratio Prediction through Segmentation Models



$$\text{volumetric\_ratio}_{RV/LV} = \frac{N_{RV}}{N_{LV}}$$

- **TotalSegmentator** nnUNet-based model trained on 1204 whole-body CTs and 620 high-resolution cardiac CTs.
- **nnUNets** trained on 13 private intermediate-expert annotations of CTPAs, with a variant initialized from TotalSegmentator weights and fine-tuned on the same private dataset.

Results

- Dice on 6 expert-annotated CTPAs.
- TotalSegmentator outperforms manual annotations, negating the need to train new nnUNets.

Nb of samples		Total Segmentator	nnUNet	Pretrained nnUNet	Inter-expert Annotations
Source domain (CT)		1824	-	1824	-
Target domain (CTPA)		-	13	13	-
Dice	LV	83.96	86.09	**87.17**	85.45
	RV	**93.47**	87.82	89.19	87.17
	Mean	**88.71**	86.95	88.18	86.31

Embolies-Arteries-Lungs Segmentation Models

4 single & multi-class nnUNets trained on 8 private intermediate-expert annotations of CTPAs.

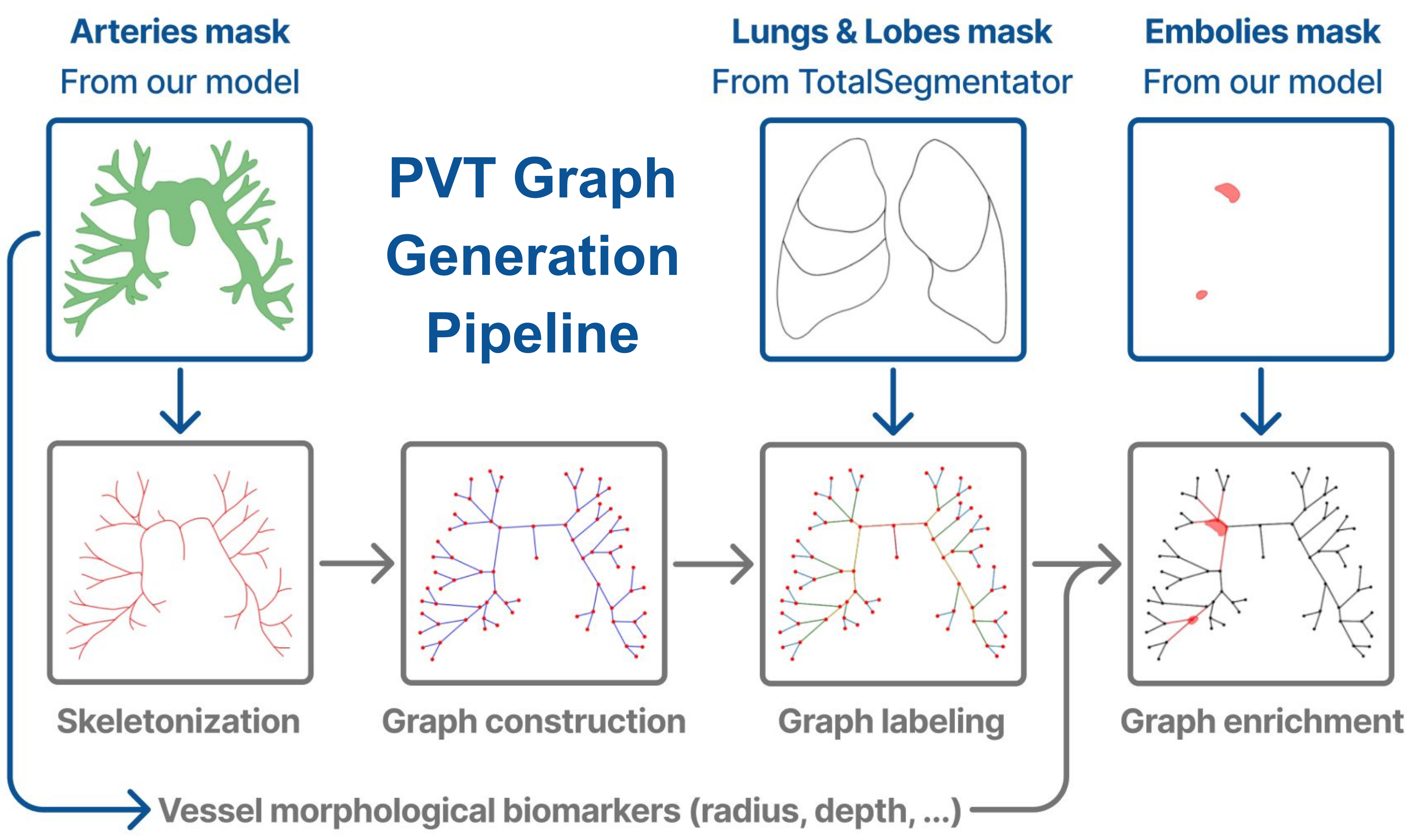
Preliminary Results

E- embolies, A- arteries, EA- embolies + arteries, EAL- embolies + arteries + lungs

		E-nnUNet	A-nnUNet	EA-nnUNet	EAL-nnUNet
Dice	Arteries	-	**90.45**	89.31	89.44
	Embolies	59.44	-	64.04	**66.79**
Avg-CC	Arteries	-	**25.54**	32.92	39.07

- Dice computed on 2 test patients for each mask, including combined classes.
- Average number of connected components (Avg-CC) is computed from 413 CTPA inferences.
- Arteries segmentation must contain a single CC.

PVT Graph Generation Pipeline



Risk Stratification Prediction through Graph-based Deep Learning Models

To predict the risk stratification, we employ a graph-level classification approach, exploring multiple GNN encoder architectures.

- **GCN** aggregates neighbor features by normalized mean.

$$h_i^{(l+1)} = \sigma\left(\text{Mean}_{j \in \mathcal{N}_i \cup \{i\}}\left(h_j^{(l)}\right) W^{(l)}\right)$$

- **GAT** aggregates neighbor features by learned attention weights.

$$h_i^{(l+1)} = \sigma\left(\sum_{j \in \mathcal{N}_i \cup \{i\}} \alpha_{ij} W^{(l)} h_j^{(l)}\right)$$

- **GIN** aggregates self and neighbor features, then applies an MLP.

$$h_i^{(l+1)} = \text{MLP}^{(l)}\left((1+\epsilon)\, h_i^{(l)} + \sum_{j \in \mathcal{N}_i} h_j^{(l)}\right)$$